Differential CC $1\mu 1\pi^{\pm}$ cross sections on argon in the NuMI beam

Updates since the collaboration meeting: releases 1.6 to 1.9

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$\nu_\mu + \bar{\nu}_\mu$ CC single charged pion on argon • FHC, RHC and combined • signal region blind, results on pseudo-data

Previous report: MicroBooNE Collaboration Meeting, Imperial College London, 12–17 September 2026

Since the collaboration meeting

Date	Change
18–19 Sep	FHC Run-1 beam-on exposure corrected (no open-trigger data): $3.283 \rightarrow 2.192 \times 10^{20}$ POT; rebuild
09 Sep–1 Oct	Control regions compared with beam-on data under a frozen protocol; corrections to the sideband simulation
26 Sep	Binning review: the 0.50 migration criterion for both families, θ_μ dropped, p_p added
18–28 Sep	New results (θ_p, $\theta_{\pi p}$, two combined-only 2D) and four status labels; validation V1
29 Sep	Version 1.6: Editorial Board package (three documents)
1 Oct	Version 1.6.1: dirt normalisation in the count and control-region macros
5 Oct	Version 1.7: Bragg-pion cut dropped (missing in the older data productions)
9 Oct	Version 1.8: detector variations applied fractionally (samples simulated with the old NuMI flux)
11–16 Oct	Version 1.9: MCS calibration of exiting muons from the control regions, hold-out test

Versions 1.7 to 1.9 tagged on 14 October. Every result is still extracted from pseudo-data; the signal region is blind.

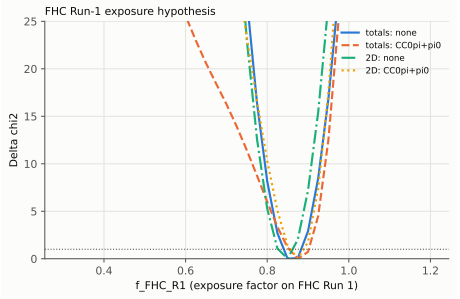
At the meeting and now

	Collaboration meeting (Draft 1.5)	Version 1.9
Exposure (10^{20} POT)	FHC 8.857, RHC 11.082, comb. 19.939	FHC 7.766, RHC 11.082, comb. 18.848
Efficiency, purity	17.4–17.6%, 57.7–58.5% (Bragg-π cut)	18.3–18.4%, 56.6–57.3% (no Bragg-π cut)
Binning criterion	diagonal ≥ 0.68 (p_μ exempt)	≥ 0.50 for every released observable
Inclusive observables	p_μ 7, $\cos\theta_\mu$ 5, θ_μ 5, $\cos\theta_\pi$ 5, $\theta_{\mu\pi}$ 5, p_π 2	p_μ 5, $\cos\theta_\mu$ 8, $\cos\theta_\pi$ 5, $\theta_{\mu\pi}$ 5, p_π 3, 2D $\cos\theta_\pi \times \cos\theta_\mu$
Inclusive totals	1.03/0.93/0.97, $\pm$37.6–38.7%	1.042/0.925/0.975, $\pm$34.1–35.9%
Proton-tagged totals	0.62/0.53/0.57, $\pm$40–47%	0.618/0.522/0.563, $\pm$40.5–42.9%
Validation	closure 1.010; 32 throws; no alternative model	closure 1.003; 100–500 members; two reweighted models
Control regions	pseudo-data only; request to open	beam-on: global 7.69/8 ($p = 0.46$); all shape tests pass
Detector variations	absolute differences	fractional (DVfrac)
MCS momentum	$\pm$5% scale term	calibration from the control regions, term `cov_MCScal`

Totals in 10^{-38} cm^2/Ar (FHC/RHC/combined), on pseudo-data.

FHC Run-1 exposure

- In the control regions FHC Run 1 sat at data/prediction 0.60–0.72; a free exposure factor fitted 0.520 ± 0.020.
- Cause: our Run-1 beam-on file holds no open-trigger data; its exposure is 2.192×10^{20} POT and 5 748 692 triggers, not 3.283×10^{20} and 9 846 635 (the open-trigger share, runs 3490–4950). Confirmed against the beam bookkeeping and the file transfer.
- Adopted: FHC $8.857 \rightarrow 7.766 \times 10^{20}$, combined $19.939 \rightarrow 18.848 \times 10^{20}$ POT; combined flux $6.60865 \rightarrow 6.59691 \times 10^{-10}$; FHC beam-off $30.2 \rightarrow 23.2$ events; about 20 hard-coded scales fixed.
- 34 universe files rebuilt. On pseudo-data the integrals move by 0.95–1.05 (the exposure cancels); on beam-on data the correction raises σ_{FHC} by 14.0% and σ_{comb} by 5.8%.
- Control-region global χ^2: $69.5/8 \rightarrow 8.1/8$.
- Still open: a free Run-1 factor fits 0.869 ± 0.022 ($\Delta\chi^2 = 37$); cause not identified.

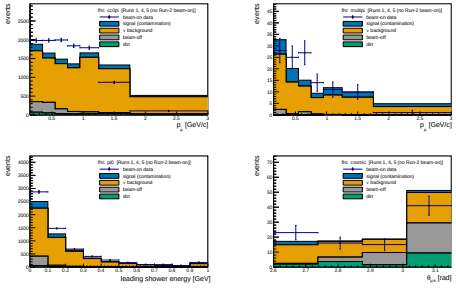


$\Delta\chi^2$ profile of the FHC Run-1 exposure factor in the control regions.

Control regions with beam-on data: frozen protocol

- Four inclusive control regions (CC0π, π^0, multi-π, cosmic), FHC Runs 1, 4, 5 and RHC Runs 1, 3, 4 (no Run-2 beam-on file); signal region excluded from the skim.
- Frozen protocol: global normalisation χ^2 (8 regions), pulls, shape tests in p_μ, $\cos\theta_\mu$ and a third variable per region.
- Corrections before interpreting: the sideband simulation predated the beam-frame fix (FHC CC0π $\cos\theta_\mu$ 350.5/4 $\rightarrow$ 3.26/4, an artefact); the sideband detector variations were not POT-normalised (detector term $\approx$ 20% $\rightarrow$ 1–2%).

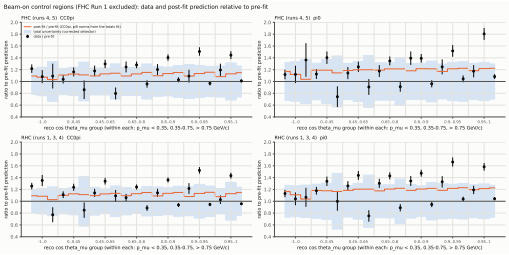
Global normalisation χ^2 (beam-on)	
first comparison (9 Sep)	8.77/8
corrected detector term, old Run-1 exposure	69.5/8
corrected Run-1 exposure	8.11/8
dirt fix (v1.6.1)	6.03/8
v1.7 selection, unchanged in v1.9	7.69/8 ($p = 0.46$)



FHC control regions, beam-on data against the prediction (v1.7 simulation, before the MCS calibration).

Control regions: background fit, constraint, far sidebands

- Joint FHC+RHC background fit: CC0π scale 0.991 ± 0.190 (v1.7); no correction warranted, FHC and RHC agree. The other periods sit 8–23% above prediction in CC0π and π^0, inside the flux and cross-section terms.
- Conditional constraint of the one-bin total from the control regions (Asimov): 35.8% $\rightarrow$ 11.3%, entirely through the flux and detector correlations. Adopting it needs a collaboration decision.
- Far sidebands (one extra cut per region): CC0π signal fraction 8.6% $\rightarrow$ 2.2% keeping 58% of the events; data/prediction CC0π 1.09–1.13, far cosmic 0.76 (the data prefer less dirt).
- The residuals are organised in p_μ: the symptom of the MCS mis-modelling (next section).

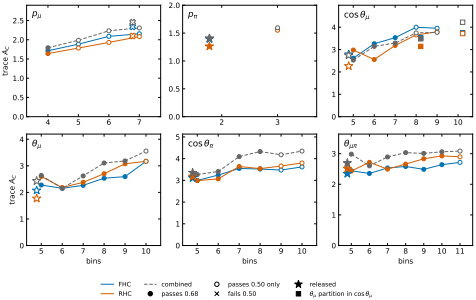


Data over prediction in $\cos\theta_\mu \times p_\mu$ groups after the fit.

Binning review: the 0.50 criterion

- 141 candidate binnings extracted in FHC, RHC and combined; maximin partition by dynamic programming, scored by the trace of A_C (degrees of freedom of the signal); the fewest bins reaching 90% of the largest admissible trace.
- The released p_μ (diagonal 0.33) and W_{had} (0.09) failed both criteria; $\cos\theta_\mu$ and θ_μ failed 0.68.
- Adopted 26 September: 0.50 for both families.

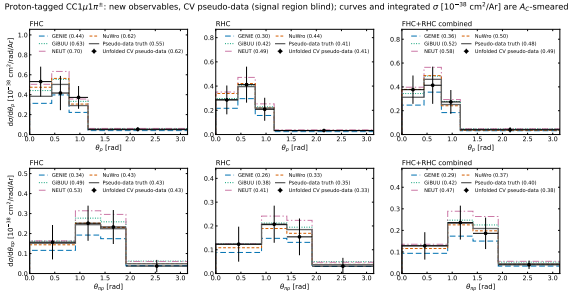
Observable	Meeting	Now
p_μ	7 bins	5 (6, then the MCS merge)
$\cos\theta_\mu$	5	8
p_π	2	3 regions (0.175, 0.205, 0.26, open)
θ_μ	5	dropped
W_{had}	6	2 (split 1.18)
$\delta\phi_T$	2	3
θ_p, $\theta_{\pi p}$	n/a, 3	4, 4
p_p	n/a	3 (0.3, 0.42, 0.60, open)



A_C degrees of freedom against the number of bins, inclusive observables.

New results and status labels

- New: θ_p and $\theta_{\pi p}$ (proton-tagged), and two 2D results in the combined configuration only, $\cos\theta_\pi \times \cos\theta_\mu$ (6 cells) and $\theta_p \times \delta p_T$ (4 cells).
- The proton candidate is the leading proton in 81.7% of events; wrong picks supply 74% of the θ_p migration.
- Four status labels, one source (`status.tsv`):
  - primary: inclusive differentials and total;
  - secondary: proton-tagged total, θ_p, $\theta_{\pi p}$, both 2D;
  - validation only: W_{had}, TKI, p_p, proton-tagged μ/π kinematics;
  - not reported: θ_μ, 5-bin p_π, $W_{\pi p}$.
- Validation V1 of the new results: widths 0.94–1.04, bias $\leq 0.79\sigma$. The release has 59 extractions.



θ_p and $\theta_{\pi p}$ on pseudo-data with the tune and four generators.

Dirt normalisation (1.6.1) and the Bragg-pion cut (1.7)

Dirt in the count macros (1 Oct)

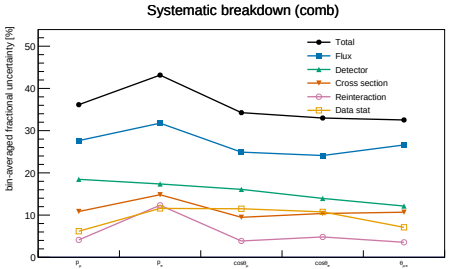
- The count and control-region macros scaled the Run-1 dirt by data POT over overlay POT and applied the 0.65 weight twice: dirt 6–21$\times$ too small. The extraction was never affected.
- Signal-region dirt 1.1/1.0/2.1 $\rightarrow$ 6.6/9.5/16.0 events; purity -0.3 points. CR dirt, FHC CC0π: 49 $\rightarrow$ 342.
- CR χ^2: pseudo-data 0.66 $\rightarrow$ 2.29/8, beam-on 8.11 $\rightarrow$ 6.03/8. A 100% dirt variation moves the totals by $\leq 0.9\%$; no dirt term carried.

Bragg-pion cut dropped (5 Oct)

- The older data productions (FHC Run 1, RHC Runs 1 and 3b) lack the Bragg-pion score: the cut was fail-open in data and applied in the simulation. The control regions showed 7–11% excess there; at unblinding the totals would have been 4.7% (inclusive) and 4.6% (proton-tagged) high.
- Cut dropped in every sample; full reprocess; 65 universe files rebuilt.
- Efficiency 17.5 $\rightarrow$ 18.3–18.4%, purity 57.3/56.6/56.9%; proton-tagged efficiency 11.5 $\rightarrow$ 12.1%.
- Totals 1.034/0.932/0.975 $\rightarrow$ 1.042/0.925/0.975; closure 0.996 $\rightarrow$ 1.003; CR global 7.69/8.

Detector variations and the old NuMI flux (1.8)

- The Run-4 detector-variation samples were simulated with the earlier NuMI flux: per POT their central value predicts 1.27 (FHC) and 1.51 (RHC) times the nominal rate, rising to 1.8–3 above 5 GeV; reweighted to the current flux, 0.94, flat.
- The absolute difference between a variation and its central value carried that ratio into the covariance. Now each variation's fractional change is applied to the nominal prediction (DVfrac).
- Detector term per bin $\times 0.70$–0.78; totals: detector 19.2–22.1% $\rightarrow$ 14.0–14.3%, total 37.7–38.2% $\rightarrow$ 34.1–35.9% (inclusive). Central values unchanged.
- Integrals $\times 1.026$ (median); combined $\cos\theta_\mu$ A_C trace 3.81 $\rightarrow$ 4.23; D'Agostini/WSVD 1.30–1.36 $\rightarrow$ 1.27–1.30.



Uncertainty breakdown per observable, combined configuration (v1.9).

MCS momentum of exiting muons (control regions)

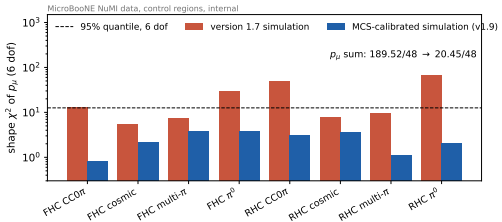
- Above 1.2 GeV/c the data hold half the predicted share of exiting muon candidates (0.157 against 0.318); data/prediction in the p_μ bins 1.00, 1.07, 1.21, 1.25, 1.09, 0.48.
- Stopping muons agree: MCS over range within 1.5% up to 3 GeV/c, so the momentum scale is fine; the exiting tracks are not.
- Cause: an angular noise of the reconstructed tracks on the scale of the MCS segments (5–11 mrad equivalent), absent in the simulation, depending on the drift-direction component and the length of the track. Only the SCE variation moves the MCS momentum, by about a fifth of the effect.
- Uncorrected impact: up to 2.7σ in a p_μ bin and +25–29% on the p_μ integral, against at most 0.65σ from the $\pm$5% term then carried.

History of the MCS term

- $\pm$5%/$\pm$10% common scale: withdrawn on 19 Sep (it cancels).
- Data-side $\pm$5% term: 0.34σ $\rightarrow$ 0.65σ through v1.8; in the official covariance from 27 Sep.
- v1.9: calibration from the control regions, term `cov_MCScal`.

MCS calibration (1.9)

- Quantile maps of the MCS momentum in 12 cells ($|\hat{d}_x|\times$ length), applied to every simulated exiting track in every sample; 19 results rebuilt, the other 40 bit-identical.
- Uncertainty: seven alternative maps in four fully correlated terms (`cov_MCScal`), replacing the $\pm5\%$ term; FHC p_μ term 6.5% (bin average), $\leq 0.28\sigma$ (FHC), 0.21σ (RHC), 0.43σ (combined).
- p_μ: top two bins merged (5 bins, last $> 0.825\,\mathrm{GeV}/c$); $\theta_p \times \delta p_T$ edge $0.8 \to 0.85$.
- Control regions: MCS χ^2 7937/24 $\to$ 27/24; p_μ data/prediction 1.00, 1.02, 1.04, 1.01, 0.98, 0.97; every shape test passes (a closure: the maps come from these events).
- Inclusive p_μ integrals against v1.8: $\times 1.110/1.018/1.071$.



Shape χ^2 of p_μ in the eight control regions, beam-on data, frozen protocol, before and after the calibration.

Hold-out test of the calibration

- Maps derived from one random half of the control-region events and tested on the other half, and vice versa: p_μ shape χ^2 21.1/5 and 5.9/5, against 7.5 ± 4.0 expected from the statistics of the halves alone; the calibrated shares agree within 6%.
- Transfers between horn modes (FHC to RHC and back) and between productions (older to newer data and back) move the top p_μ bin by 3–7%, the spread of the period-group maps that build the uncertainty.
- The calibration therefore transfers between independent events, horn modes and productions within its quoted uncertainty; the closure of the previous slide is not the only evidence.
- Record: `report/results/mcs_cal_holdout.{md,json}`, `scripts/mcs_cal_holdout.py`.

Validation

Closure and ensembles

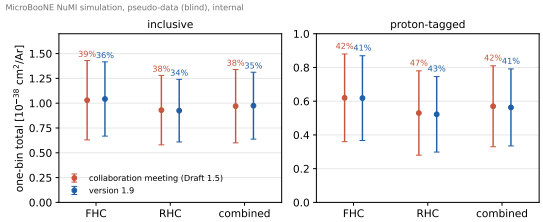
- Closure: inclusive $\chi^2 \leq 3.12/8$, ratios 0.905–1.106, mean 1.003 (meeting: 1.010); proton-tagged $\chi^2 \leq 7.28/4$.
- Ensembles: 32 throws at the meeting, then six 100-member ensembles (20 Sep), 500 members for the binning review, all rebuilt for v1.7 to v1.9. Widths 0.92–1.04 (inclusive), 0.89–1.04 (proton-tagged); offsets within 1.4% at $\leq 1.5\sigma$.
- 24 Sep: concurrent arrays re-threw shared files and ten members read truncated pseudo-data (RHC total width $1.28 \to 0.99$ after the fix); throws are now created once (`link(2)`).

Alternative models

- Pseudo-data from GENIE multisim universe 545 and the $\Delta \to N\pi$ angular unisim, extracted with the nominal response: largest inclusive bias 1.21σ (FHC $\cos\theta_\mu$ bin 3), RHC $\leq 0.90\sigma$; proton-tagged low $\delta p_T/p_n$ $-1.19/-1.02\sigma$, lower W_{had} $+1.04\sigma$.
- Since v1.7 the truth changes come from the universe ratios, not from two separate throws.
- NuWro closure dropped (19 Sep): the Run-1 FHC NuWro overlay has a signal rate $1.76\times$ GENIE against 1.6 expected and its production could not be traced. Independent-generator closure (U3) still pending.

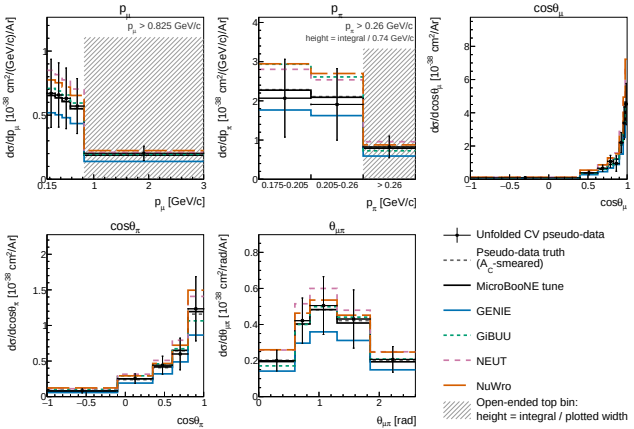
One-bin totals (1.9)

10^{-38} cm^2/Ar	Inclusive	Proton-tagged
FHC	1.042 ± 0.374 (35.9%)	0.618 ± 0.251 (40.6%)
RHC	0.925 ± 0.316 (34.1%)	0.522 ± 0.224 (42.9%)
Combined	0.975 ± 0.337 (34.6%)	0.563 ± 0.228 (40.5%)



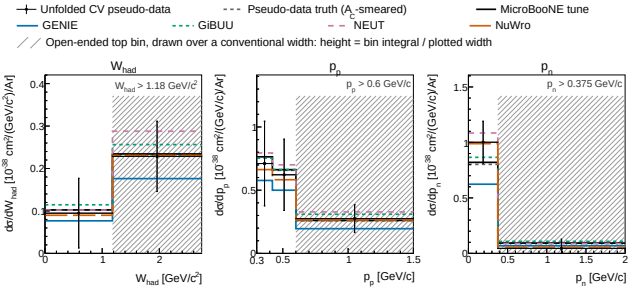
- FHC inclusive: flux 30.4%, detector 14.2%, cross section 10.6%, reinteraction 3.8%, POT 3.4%, targets 1.7%, data statistics 4.4%.
- Selected / background: 1522/651, 1881/817, 3404/1467; proton-tagged 677/328, 834/420, 1510/749.
- FHC generators: GENIE 0.82, GiBUU 1.13, NEUT 1.32, NuWro 1.25.
- Against the meeting: central values within 1%; uncertainty $\approx 38\% \rightarrow$ 34–36% (DVfrac). On beam-on data the Run-1 correction adds +14% to FHC.

Inclusive differential results (1.9)



- Combined configuration, unfolded pseudo-data with the A_C-smeared truth, the tune and four generators.
- p_μ 5, $\cos\theta_\mu$ 8, $\cos\theta_\pi$ 5, $\theta_{\mu\pi}$ 5 bins, p_π 3 regions.
- Integrals (FHC/RHC/comb.): p_μ 0.97/0.70/0.86, p_π 0.84/0.74/0.78, $\cos\theta_\mu$ 0.72/0.71/0.75, $\cos\theta_\pi$ 0.66/0.62/0.67, $\theta_{\mu\pi}$ 0.89/0.81/0.84 (meeting 0.62–0.96).
- Bin-averaged systematic total 31–49% (meeting: 32–54% total).

Proton-tagged results (1.9)



- Combined configuration on pseudo-data: W_{had} 2 bins (split 1.18 GeV), TKI 2–3 bins (δp_T 2, $\delta\alpha_T$ split at 120°, $\delta\phi_T$ 3, p_n 2), p_p 3, θ_p 4, $\theta_{\pi p}$ 4.
- W_{had}, TKI and p_p are validation-only; the proton-tagged total, θ_p, $\theta_{\pi p}$ and the $\theta_p \times \delta p_T$ 2D result are secondary.
- Open: inverted-proton-PID control region (P1), wrong-candidate term for θ_p (P2), model sensitivity of the low $\delta p_T / p_n$ bins (P3), W_{had}/p_n reconstruction validity (P4).

Documents and process

- Three documents since 19 Sep: inclusive analysis note (approval path), proton-tagged note (validation), shared technical supplement; cross-referenced, one preamble.
- Review responses (27 Sep): status vocabulary, official covariance with the MCS term, `check_documents.py` for consistency; approval material only in `approval_request.tex`.
- Release packages with notes in `report/releases/release_v1.*`; v1.6 for the Editorial Board (29 Sep), v1.6.1 errata (1 Oct), v1.7–v1.9 tagged 14 Oct.
- The documents of v1.6 to v1.9 describe the analysis before the control regions; 18 beam-on passages are held out until the control-region comparison is approved.
- PRD/PRL paper drafts started on 19 Sep, on hold since 15 Oct.

Open items

Before unblinding

- U1: written confirmation from the NuMI flux group that the flux constants are the PPFX-corrected central value.
- U2: control-region comparison under the frozen procedure approved; convener ratification of the procedure and the π^0 target; second-person skim verification.
- U3: independent-generator closure with a validated NuWro overlay.
- U5: beam-on POT and triggers per file checked against the beam database to $< 1\%$.
- U6: clean-cache reproduction of the release to 10^{-9}.
- FHC Run-1 residual: $13 \pm 2\%$ ($f = 0.869 \pm 0.021$), cause not identified.

Decisions and later

- R1: collaboration decision on the combined-only 2D results.
- Adoption of the control-region conditional constraint (35.8% $\rightarrow$ 11.3% on the total, Asimov).
- U7 (before publication): beamline-geometry flux term, beam-off gate-ratio term and the Run-2 beam-off stand-in term in the covariance.
- Proton-tagged P1–P4.
- Ensemble offset: the six inclusive 1D means are below the reference ($\leq 0.9\%$, $\leq 1.5\sigma$); the 2D $\cos\theta_\pi \times \cos\theta_\mu$ is 0.8% low at 2.4σ.
- U4 (MCS) complete with v1.9.

Summary

- Four releases since the meeting. The central values on pseudo-data moved by less than 1%; the total uncertainty fell from about 38% to 34–36% (inclusive) through the fractional detector variations.
- Exposure and selection corrected where the beam-on control regions pointed: the FHC Run-1 exposure (+14% on σ_{FHC} for data), the Bragg-pion cut missing in the older productions (4.7% bias at unblinding avoided), the dirt in the count macros.
- Binnings now follow one criterion (0.50) for every released observable; new θ_p, $\theta_{\pi p}$ and two 2D results, with status labels.
- The MCS momentum of exiting muons is calibrated from the control regions: every control-region shape test passes, the hold-out test transfers within the quoted term.
- Control regions with beam-on data: global 7.69/8 ($p = 0.46$). Open before unblinding: the flux constants, the approval of the control-region procedure, an independent-generator closure, the Run-1 residual.

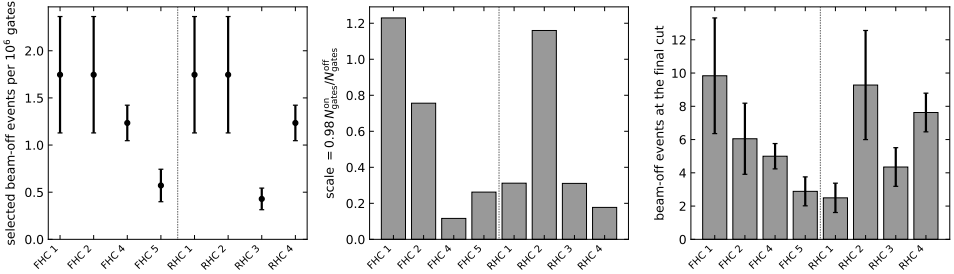
Backup

Backup: control-region shape tests with the calibrated simulation

Mode	Distribution	v1.7	calibrated
FHC	CC0π p_μ	13.29/6	0.84/6
FHC	cosmic p_μ	5.36/6	2.14/6
FHC	multi-π p_μ	7.55/6	3.77/6
FHC	π^0 p_μ	29.56/6	3.83/6
RHC	CC0π p_μ	50.01/6	3.10/6
RHC	cosmic p_μ	7.83/6	3.58/6
RHC	multi-π p_μ	9.69/6	1.14/6
RHC	π^0 p_μ	66.23/6	2.05/6
sum	p_μ	189.52/48	20.45/48
sum	other distributions	42.52/66	42.52/66

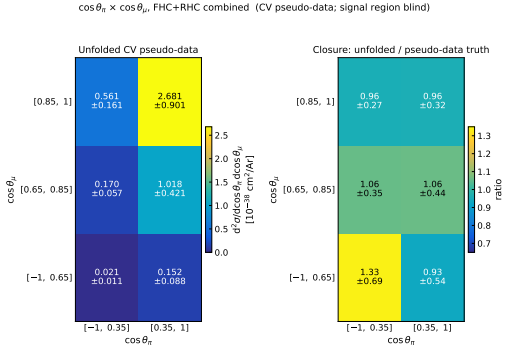
- Beam-on data, frozen protocol, from `report/results/cr_protocol_mcscal.md`.
- The global normalisation χ^2 is unchanged (7.69/8): the calibration moves events between p_μ bins, not between regions.
- Pseudo-data with the calibrated simulation: p_μ sum 22.9/48, the reference for the data.

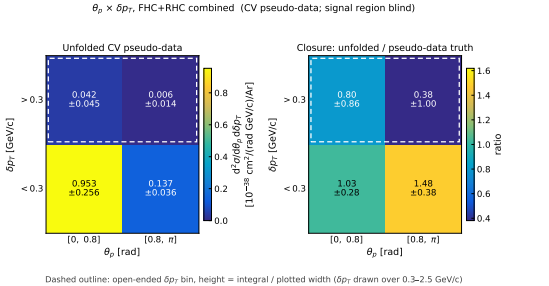
Backup: beam-off per run period



Beam-off rate per run period with the corrected Run-1 trigger count.

Backup: two-dimensional results (combined)





$\cos\theta_\pi \times \cos\theta_\mu$ (6 cells, inclusive) and $\theta_p \times \delta p_T$ (4 cells, proton-tagged), unfolded pseudo-data and the tune; combined configuration only (R1).

Backup: uncertainty breakdown, FHC and RHC

